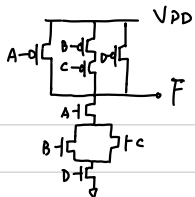


HW1
$$F = \overline{A + B\overline{C} + D}$$

$$= \overline{A \cdot (\overline{B\overline{C}}) \cdot D}$$

$$= \overline{A(B+C)D}$$



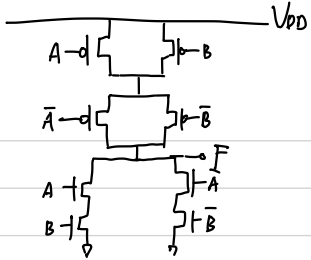
$$F = A \oplus B$$

$$= A\overline{B} + B\overline{A}$$

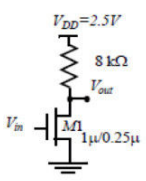
$$= \overline{(A\overline{B})(B\overline{A})}$$

$$= \overline{(A+B)(\overline{B}+\overline{A})}$$

$$= \overline{A\overline{B} + B\overline{A}}$$



(二) 如图所示电路, 已知 M1 的 $V_{th0}=0.43V$, $V_{DSAT}=0.63V$, $K'=115 \times 10^6 A/V^2$, 沟道尺寸如图所示, 单位沟道宽度的覆盖电容 $C_{gs0}=C_{gdo}=0.31fF/\mu m$, 单位面积的氧化层电容 $C_{ox}=6 fF/\mu m^2$.



- 1) 请计算 (忽略体效应和沟道长度调制效应): $V_{in}=2.5V$ 时 V_{out} 的稳态电压(记为 V_{OL})是多少? M1 管处在什么工作区?
- 2) 如果 V_{in} 从 0V 上升到 2.5V, 而 V_{out} 的初始电压等于 2.5V, 那么从 I_n 端变化开始到 V_{out} 达到稳态这个过程中由 I_n 端注入的总净电荷量等于多少?

1) $V_{in}=2.5V$ M1, $V_{gs}=2.5V$ $V_{ds}=V_{out}$

若 $V_{gs}-V_t > V_{ds}$

$$\begin{cases} V_{DD} - I_D R = V_{DS} \\ I_D = k_n' \frac{W}{L} \left[V_{gt} V_{ds} - \frac{V_{ds}^2}{2} \right] \end{cases}$$

$$\Rightarrow \frac{V_{DD} - V_{DS}}{R} = k_n' \frac{W}{L} V_{gt} V_{ds} - k_n' \frac{W}{L} \frac{V_{ds}^2}{2}$$

$$\Rightarrow \frac{1}{2} k_n' \frac{W}{L} V_{ds}^2 + \left(k_n' \frac{W}{L} V_{gt} - \frac{1}{R} \right) V_{ds} + \frac{V_{DD}}{R} = 0$$

$$\Rightarrow V_{DS} = 0.31V$$

2) $Q = CV$ 逐段讨论 C 的情况 $C = C_{ox} + C_{gs} + C_{gd}$

① $V_{in} < V_t$ $C_1 = C_{ox} + C_{gs} + C_{gd} = 6 \times 0.25 \times 1 + 0.31 \times 1 \times 2 = 2.12 fF$ $V < 0.4V$
 ② $V_{in} > V_t$ $V_{in} - V_t < V_{DSAT}$ $C_2 = \frac{2}{3} C_{ox} + C_{gs} + C_{gd} = 1.62 fF$ $0.4V < V < 1.06V$
 ③ $V_{in} - V_t > V_{DSAT}$ $C_3 = C_{ox} + C_{gs} + C_{gd} = 2.12 fF$ $1.06V < V \leq 2.5V$

$$\therefore Q = \int_0^{V_{DD}} C dV = 2.12 \times (0.4) + 2.5 - 1.06 + 1.62 \times (1.06 - 0.4) = 4.985 fC$$

(三) 一个 CMOS 反相器, 已知 NMOS 管的 $W/L=0.1\mu/0.065\mu$, 阈值电压等于 0.4V, $W/L=1$ 的 NMOS 管 V_{GS} 等于阈值电压时的漏源电流等于 100nA; PMOS 管的 $W/L=0.2\mu/0.065\mu$, 阈值电压等于 -0.4V, $W/L=1$ 的 PMOS 管 V_{GS} 等于阈值电压时的漏源电流等于 50nA; NMOS 管和 PMOS 管的亚阈值耦合系数都是 $n=1.4$. 电源电压等于 1V, 当输入电压稳定为 0V 时, 请计算此稳态条件下室温下的电源电流 (仅考虑亚阈值漏电流)。当工艺改变, 使 n 变成 1.3 (其它参数不变), 静态电源电流会如何变化?

$kT/q = 25.85m$

$$I_D = I_0 e^{\frac{V_{GS}-V_T}{nK_T/q}} (1 - e^{-\frac{V_{DS}}{K_T/q}})$$

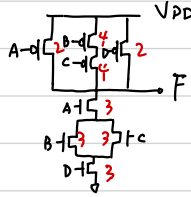
 for N $V_{GS} = 0.4V$ $V_{DS} = 0V$ $I_0 = I_0 = 100nA$
 for P $V_{GS} = -0.4V$ $V_{DS} = 0V$ $I_0 = I_0 = 50nA$

for $\frac{W}{L} = \frac{0.1}{0.065}$ $I_{n0} = 100n \cdot \frac{0.1}{0.065} = 153.85nA$ $I_{p0} = 50n \cdot \frac{0.2}{0.065} = 153.85nA = I_{n0}$

$$\therefore V_{in} = 0V$$
 PMOS 导通 PMOS $V_{GS} = 0V$ $I_D = I_{p0} e^{\frac{0-0.4}{-1.4 \times 25.85m}} = 153.85n e^{\frac{-0.4}{-36.19m}} = 2.6 fA$
 $n=1.3$ $I_D = 153.85 \exp(-0.4 / (1.3 \times 25.85m)) \approx 0.0011nA < 0.0016nA$ $\therefore$ 变小

I. Please draw the transistor level circuit of the compounded static complementary gate that implements the function $F = \overline{A + BC + D}$ and annotate the scale of the transistors. And then, please calculate the logic effort for each input and the normalized intrinsic delay of the gate.

$$F = \overline{A + BC + D} = \overline{A + (B+C)D}$$



Scale: NMOS $M_A = M_B = M_C = M_D = 3$

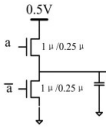
PMOS $M_A = M_B = 2$ $M_C = M_D = 4$

$g_A = 5/3$ $g_B = 7/3 = g_C$ $g_D = 5/3$

$P = (4+3+2+2)/3 = 11/3$

$d = P + \frac{g_f}{f}$

II. The following figure is the schematic of a kind of low-swing bus driver. Known that: for the NMOS FETs $V_T = 0.43V$, $V_{DSAT} = 0.63V$, $k' = 115 \times 10^{-6} A/V^2$; the channel sizes of all transistors are as shown; the input swing is 2.5V, the supply voltage of the driver is 0.5V, the total load capacitance is 1pF, please calculate the rise propagation delay and the fall propagation delay of it. (Ignore the body effect and the channel length modulation effect and Calculate equivalent driving resistance of the gate as the average of the value of it at the initial and the end final states.) (Note: the two transistors are both NMOSFET!)



考虑 $t_p = 0.69 R_{eq} C_L$ 其中 $R_{eq} = \frac{1}{2} (R_{on初} + R_{on末})$

∴ 分别计算初末态 R_{on} 即可

由于 $V_{ds} \leq V_{DD} < V_{DSAT}$ 如果 MOS 导通则 必在线性区

① 初态 $V_{in} = 2.5V$ 对 M_1 $V_{DS} = 0.5V$ $V_{GS} < V_{gs}$ $V_{ds} < V_{DSAT}$

线性区 $R_{on1} = \frac{V_{DS}}{k'E(V_{GS} - V_T)V_{DS}} = 1.194 k\Omega$

② 末态 $V_{in} = 2.5V$ 对 M_1 $V_{DS} = \frac{1}{2} \cdot 0.5V$ (在线性区, 平均电压)

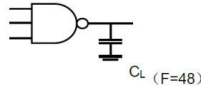
∴ $V_{DS} = 0.25V$ $V_{GS} = 2.5 - 0.25V = 2.25V$ $R_{on2} = 1.289 k\Omega$

最后用 $t_p = \frac{1}{2} (R_{on1} + R_{on2}) C_L = 0.86 ns$

fall 初态 $V_{in} = 0V$ $V_{GS} = 2.5V$ $V_{DS} = 0.5V$ 末 $V_{in} = 0V$ $V_{GS} = 2.5V$ $V_{DS} = 0.25V$

一、一个 3 输入 NAND 门驱动一个电容, 等效扇出为 48, 已知自载系数为 1, 要对该电路进行延时优化。

(1) 请计算说明应该插入几级反器



(2) 如果第一级门的尺寸系数为 1, 请给出最小延时时所对应的各级逻辑门的尺寸系数; 计算对应的最短归一化路径延时。

(2) $h = \sqrt[3]{F} = \sqrt[3]{48} \approx 4.109$

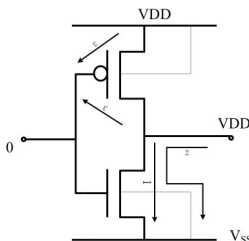
II $S_2 = g_1 f_1 / g_2 \cdot f_1 = g_1 f_1 = h = 4.109$

I $g_1 = \frac{1}{3}$ $f_1 = h/g_1 = 2.5854$ $S = 1$

III $S_3 = \frac{g_2 f_2}{g_3} \cdot f_1 f_2 = g_1 f_1 h/g_2 = h^2 = 18.56$

$D = NH^{\frac{1}{2}} + N - n_i + P = 3\sqrt{48} + 3 - 1 + 1/3 = 17.92$

二、考察一个输入为 0 的静态 CMOS 反相器, 请列出造成其静态功耗的漏电流分量和其对应的物理效应名称。



- 1 nmos 亚阈值漏电
- 2 nmos GIDL+漏区 PN 结 BTBT 漏电
- 3 PMOS 栅氧隧穿漏电

综合一个由随机逻辑构成的芯片, 逻辑的平均活动因子为 0.1。

采用平均等效翻转电容为 450pF/mm² 的标准单元工艺。如果你的芯片的面积为 70mm², 在 $V_{DD} = 0.9V$ 下的运行频率为 450MHz,

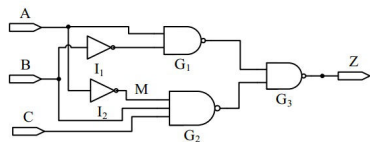
试估算它的动态功耗。

$$P_{dy} = C_L S V_{DD}^2 f \approx 0.1$$

$$= 450p \cdot 70 \cdot 0.1^2 \cdot 450M \cdot 0.1$$

$$= 1.15 W$$

四、 下图的组合逻辑电路的所有输入端的信号同时到达，假设为为了在不损失性能的前提下降低静态功耗，我们采用多阈值电压工艺的标准单元库来实现，即库中除了常规器件构成的逻辑门，还提供了高阈值电压器件构成的逻辑门单元。那么下面各个逻辑门中哪几个单元可以考虑采用高阈值电压器件单元？请说明理由。



想在不损失性能前提下降低静态功耗

① 依找出关键路径

② 延时随 V_T 上升而增大

静态功耗随 V_T 上升而减小

③ 关键路径上单元不变，而非关键路径

应使用高阈值单元

$\therefore V_T \uparrow \begin{matrix} t_p \uparrow \\ \text{Power} \downarrow \end{matrix} \Rightarrow \begin{matrix} \text{关键路径不变性能不变 } t_p \\ \text{非关键路径 power} \downarrow \end{matrix}$

路径与旁 ① $A \rightarrow G_1 \rightarrow G_3 \rightarrow Z$ ② $A \rightarrow I_2 \rightarrow G_2 \rightarrow G_3 \rightarrow Z$ ③ $B \rightarrow I_1 \rightarrow G_1 \rightarrow G_3 \rightarrow Z$ ④ $B \rightarrow G_2 \rightarrow G_3 \rightarrow Z$ ⑤ $C \rightarrow G_2 \rightarrow G_3 \rightarrow Z$

$\therefore$ ① $t_{a1} + t_{a3}$ ② $t_{i2} + t_{g2} + t_{g3}$ ③ $t_{i1} + t_{g1} + t_{g3}$ ④ $t_{g2} + t_{g3}$ ⑤ $t_{g2} + t_{g3}$

其中 $t_{g2} > t_{a1} = t_{a3} > t_{i1} = t_{i2}$ $\therefore$ ② 是关键路径

$\therefore I_2, G_2, G_3$ 不可用, I_1, G_1 可用